Patrick Selby

614-332-6860 ● patrickselby256@gmail.com ● linkedin.com/in/pat-selby ● github.com/pat-selby

EDUCATION

Grambling State University Expected May 2028

B.S. in Cybersecurity, Minor in Computer Information Systems | GPA: 3.88/4.00

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Security

WORK EXPERIENCE

CodePath | Software Engineering Fellow February 2026 – May 2026

- Built a security-log processing pipeline to automate event intake, using Python and PostgreSQL to ingest 150K+ unstructured logs and reduce ingestion time by 85%.
- Architected multithreaded regex analysis across 50K concurrent event streams, cutting intrusion-detection latency by 35% in the security-event pipeline.
- Debugged and validated 14 backend modules with pytest, achieving 95% branch coverage and catching 14 critical bugs before deployment.

AI4ALL Ignite | Machine Learning Research Fellow June 2026 – August 2026

- Investigated language-model inference performance by profiling KV-cache overhead and dynamic batching, identifying 3 bottlenecks and reducing response latency by 25%.
- Built PyTorch/CUDA harnesses for multi-head attention, increasing tensor-core utilization by 28% under high concurrency.
- Benchmarked quantization, kernel fusion, KV-cache compression, and dynamic batching with a four-person team, reducing GPU memory usage by 34% and increasing throughput 2.4× in a co-authored technical report.

PROJECTS

UniQR | Python, C++20, OpenCV, pytest

- Built UniQR to read on-screen QR codes without a phone, decoding 96.8% of 1,064 damaged codes across 7 formats with zero wrong decodes; shipped 20 releases with pytest and GitHub Actions CI.
- Reduced median QR scan time from 115 to 5 ms by porting the scanner and Windows app to C++20 without OpenCV, preserving near-identical accuracy and zero wrong decodes across 1,064 test codes.

PacketShield | C++20, CUDA, CMake, CTest

- Built PacketShield to evaluate GPU acceleration for network packet inspection, using C++20/CUDA signature-matching kernels, pinned memory, and asynchronous transfers to process 100K-packet batches.
- Benchmarked the engine on an NVIDIA T4, reducing batch runtime from 429.5 to 32.1 ms against a multithreaded CPU baseline while preserving 100% CPU/GPU verdict agreement on the synthetic test batch.

Kairo: AI Research Workspace

Team stack: React, Vite, Go, PostgreSQL/Neon, Python, Docker, ElevenLabs

- Refined Kairo, an AI workspace for planning and running research experiments, with React fixes across 3 areas; verified 91 frontend tests, a clean build, and `go vet`, documenting 2 local backend test failures.
- Prepared a manual-plan demo that completed an experiment against live PostgreSQL in about 18–19 seconds with 13 evidence files, contributing to a 4-person team's 2nd-place finish at Grambling's Hacktoberfest Hack Day.

LEADERSHIP & ACTIVITIES

TigerHacks | Organizer August 2026

- Led a 60-participant hackathon, securing 4 sponsors, $3,500 in funding and prizes, and 8 industry mentors.

ColorStack Grambling | Events Committee Member August 2025 – Present

- Co-organized 4 career events reaching 200+ students and reviewed 40 peers' resumes, with 65% securing tech internship interviews.

Grambling Cybersecurity Alliance | Technical Development Chair January 2026 – Present

- Organized 18 Hack The Box and TryHackMe prep sessions; all 3 competitors placed in the top 10%, and the team was selected for DOE's 2026 CyberForce Competition.

SKILLS

Languages: Python, C++20, TypeScript, SQL
Frameworks & Libraries: React, PyTorch, OpenCV, CUDA
Databases: PostgreSQL
Developer Tools: Git/GitHub, GitHub Actions (CI), pytest, CMake, CTest